

Endothelial adhesion molecule ESAM binds directly to the multidomain adaptor MAGI-1 and recruits it to cell contacts

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Abstract

Endothelial cell-selective adhesion molecule (ESAM) is an immunoglobulin-like transmembrane protein associated with endothelial tight junctions (TJ). Based on a yeast two-hybrid screen, we have identified the membrane-associated guanylate kinase protein MAGI-1 as an intracellular binding partner of ESAM. MAGI-1 is a multidomain adaptor protein, which binds to transmembrane, cytoskeletal, and signaling molecules, and has been localized to tight junctions in epithelial cells. MAGI-1 associates with the very C-terminal sequence of ESAM most likely through a PDZ domain-mediated interaction. The direct interaction between ESAM and MAGI-1 was confirmed by pull-down experiments. The two proteins formed stable complexes in transfected Chinese hamster ovary (CHO) cells, which could be immunoprecipitated. We found MAGI-1 to be associated with cell–cell contacts in human umbilical vein endothelial cells (HUVECs) and in mouse endothelium, where it colocalizes with ESAM. In CHO cells, recruitment of MAGI-1 to cell contacts required the presence of ESAM. Hence, ESAM may be involved in anchoring MAGI-1 at endothelial tight junctions.

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Introduction

Endothelial cells play essential roles in the control of vascular permeability for solutes and macromolecules, as well as in the regulation of leukocyte diapedesis. In addition, they mediate the formation and outgrowth of blood vessels. These functions are dependent on the formation and the disintegration of interendothelial cell junctions that are composed of different highly specialized and regulated structures, for example, the tight junctions (TJ) and the adherens junctions (AJ) [1,2]. AJs confer physical strength to the tissue and are involved in morpho-

genesis [3], whereas the TJs seal the cell layer for solutes and macromolecules.

Each type of junctional structure is composed of specific protein complexes with one or several integral membrane proteins and peripheral membrane components associated at the cytoplasmic plaque. Adhesion molecules localized to the AJs, such as the vascular endothelial cadherin (VE-cadherin), hold apposing endothelial cells together while their cytoplasmic tails organize actin-based cytoskeletal supports. It has been shown that VE-cadherin is directly involved in the maintenance of endothelial cell contact integrity [4]. TJ molecules, such as occluding [5] and claudins [6], form continuous strand structures that fulfil two main functions: first, they restrict the lateral diffusion of lipids and integral transmembrane proteins between the apical and basolateral cell compartment, and thereby sustain the polarity of endothelial cells [7]. Second, TJs seal the endothelial cell layer, limiting the paracellular permeability

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for ions, solutes, and macromolecules in a tissue-specific manner [8].

Apart from occludin and claudins, there have been identified several immunoglobulin (Ig) superfamily members as integral membrane proteins at TJs. These include junctional adhesion molecules (JAMs) [1], endothelial cell selective adhesion molecule (ESAM) [9,10], the coxsackievirus and adenovirus receptor (CAR) [11], and the recently published CAR-like membrane protein (CLMP) [12]. All of these belong to the CTX subset of the Ig superfamily, which is characterized by one V-like and one C2-like extracellular Ig domain [13]. In contrast to claudins, ectopic expression of JAM-A (formerly designated JAM-1, nomenclature according to Ref. [14]) does not lead to the formation of TJ strands [15], suggesting that the function of JAM-A, and probably the function of the other CTX family members at TJs as well, is different from the function of claudins. JAM-A, -B, and -C directly bind to the cell polarity protein PAR-3 [15–17], which is part of a ternary protein complex consisting of PAR-3, PAR-6, and the atypical protein kinase C (aPKC). This molecular complex is crucial for the formation of TJs and the constitution of cell polarity [18]. The interactions between the JAMs and PAR-3 might target the whole complex to the TJs, indicating that JAMs might play a pivotal role in junction formation and polarity establishment. The role of the other Ig-superfamily members at TJs is less clear.

Insight into the putative function of integral membrane proteins is often gained from the identification of proteins associated with the cytoplasmic domain. A newly identified interaction partner of the CTX family member CAR is the ligand-of-numb protein X (LNX), which is bound in a PDZ domain-dependent manner [19], and was previously shown to bind to and regulate the expression level of the cell-fate determinant Numb [20,21]. Therefore, CAR might be involved in the formation of a functional complex containing LNX and Numb. A large family of potential CTX protein ligands at TJs is the membrane-associated guanylate kinases (MAGUKs). They are multidomain proteins comprising a catalytically inactive guanylate kinase domain (GuK), a SH3 domain, and several PDZ domains. All of these domains are potential sites of protein–protein interaction, indicating that MAGUKs might act as adaptor proteins forming the interface between transmembrane and cytoplasmic proteins. For example, the MAGUK protein ZO-1 interacts with JAM-A, -B, and -C [16,22,23], and is recruited directly or indirectly to sites of cell–cell contact by the coxsackievirus and adenovirus receptor (CAR) in nonpolarized cells [11].

A particular subset of the MAGUK protein family is the membrane-associated guanylate kinase with inverted domain structure (MAGIs). They are composed of one amino terminal GuK domain, two WW domains, which replace the SH3 domain in other MAGUKs, and five to six PDZ domains depending on the isoform and splice variant. The first of three so far discovered MAGI isoforms (MAGI-1)

was identified as an interaction partner of murine K-RasB [24]. Human MAGI-1 was shown to bind to the brain angiogenesis inhibitor 1 (BAI1) and was therefore named BAI-associated protein 1 (BAP1) [25]. The protein is localized at TJs in most epithelial cell types and seems to be expressed in endothelial cells [26]. There are several cytoplasmic interaction partners of MAGI-1 known so far, for example, beta-catenin [27], MuSK [28], synaptopodin, and α -Actinin-4 [29]. The interaction of MAGI-1 with the actin-bundling molecules synaptopodin and α -Actinin-4 indicates that MAGI-1 might be implicated in actin cytoskeleton dynamics in polarized epithelial cells. Besides BAI1, which is a putative seven-span transmembrane protein, there have been identified only two further transmembrane ligands of MAGI-1: Megalin is an endocytosis receptor that is expressed exclusively on the apical surfaces of epithelial cells [30,31], whereas JAM4 is a new CTX family protein that is expressed at junctions of epithelial cells, but has not been described on endothelial cells [32].

ESAM is a recently identified CTX family member that is located at TJs [9,10]. It is similar to other CTX molecules like the JAMs, but is more closely related to CAR, CLMP, and BT-IgSF, the brain- and testis-specific immunoglobulin superfamily member [33]. Like these closely related proteins, it contains a C-terminal type I PDZ targeting motif, whereas the JAMs, except JAM4, possess a type II motif. Interestingly, ESAM has been found exclusively on endothelial tissues, except for megakaryocytes and platelets, which strongly express the protein. ESAM-deficient mice are viable and phenotypically normal, but they show decreased tumor growth, tumor angiogenesis, and neo-vascularization in matrigel plug assays [34].

To gain more insight into the putative function of ESAM, we have tried to identify intracellular ligands of the protein. We found that MAGI-1 associates directly with the TJ molecule ESAM in a PDZ domain-dependent manner. In addition, we found that MAGI-1 is indeed expressed at cell contacts between human umbilical vein endothelial cells (HUVECs) and between mouse endothelial cells. Localization studies in transfected Chinese hamster ovary (CHO) cells revealed that MAGI-1 was only recruited to cell contacts if ESAM was coexpressed. Our data suggest that the interaction between ESAM and the multidomain adaptor protein MAGI-1 might serve to recruit MAGI-1 to TJs and thus contribute to the formation of large molecular complexes at TJs in endothelial cells.

Materials and methods

Cell culture

The murine endothelioma cell line bEnd.3 derived from mouse brain capillaries [35] was provided by Dr. Werner Risau (Max-Planck-Institute for Physiological and Clinical Research, Bad Nauheim, Germany). Cells were maintained

in Dulbecco's modified Eagle's medium supplemented with 10% fetal calf serum, 2 mM L-glutamine, and penicillin/streptomycin (Life Technologies, Inc., Eggenstein, Germany). CHO dhfr⁻ cells were maintained in α -minimal essential medium supplemented with 10% fetal calf serum, 2 mM L-glutamine, and penicillin/streptomycin. Stable transfected CHO dhfr⁻ cells expressing full-length ESAM were cultured as described [9]. CHO dhfr⁻ cells stably expressing C-terminally truncated ESAM were generated by transfection with pcDNA3-ESAM/ Δ 5 as described [9]. Human umbilical vein endothelial cells (HUVECs) were isolated and cultured according to standard procedures [36].

Expression vectors

Expression vector inserts were generated by polymerase chain reaction (PCR) from plasmid DNA or from first strand cDNA. First strand cDNA was generated from total RNA derived from the mouse endothelioma cell line bEnd.3 using standard techniques [37]. Yeast expression vectors were constructed in pBTM116 (bait plasmids) and pVP16 (prey plasmids), thereby generating fusion proteins between LexA and ESAM or between the VP16 activation domain and the prey protein, respectively. The ESAM bait vector (pBTM116-ESAM) was generated by subcloning the cytoplasmic tail (amino acids 275–394) of mouse ESAM into pBTM116. The insert sequence of this construct was PCR amplified from the cloned ESAM cDNA [9] using the forward primer 5'-GGAATTC-CAGCGCCGGAGCAAGACCTTG-3' and the reverse primer 5'-TGGATCCTCACACAAGAGACCCAGCCTG-3', digested with *Bam*HI and *Eco*RI and cloned into the equally restricted vector pBTM116. For eukaryotic expression, cDNA fragments were cloned into pKE081-myc [23], a vector including a Kozak consensus sequence for translation initiation, allowing expression of cDNA fragments representing internal protein sequences. Additionally, pcDNA3.1/V5-HisTOPO (Invitrogen BV, Groningen, The Netherlands) and pCMV-Tag4A (Stratagene) were used for eukaryotic expression. The pVP16 insert sequences of positive yeast clones were subcloned into pKE081myc. This led to the generation of pKE081myc-MAGI-1_{792–901}, encoding amino acids 792–901 of mouse MAGI-1 (GenBankTM accession number AF027503). The newly identified MAGI-1 splice variant MAGI-1 α β 2 γ was PCR amplified from bEnd.3 cDNA using the forward primer 5'-CCGCCATGTCGAAAGTGATCCAGAA-3' and the reverse primer 5'-TCAGATGCTGAGGTCGGTG-3', cloned into pcDNA3.1/V5-HisTOPO, and sequenced. In this full-length construct, the endogenous stop codon is included, leading to the expression of the respective protein without V5-His tag. The MAGI-1 α β 2 γ cDNA was subcloned into pCMV-Tag4A thereby generating MAGI-1 α β 2 γ -FLAG (for brevity, called MAGI-1-FLAG). The expression vectors encoding full-length ESAM (pcDNA3-ESAM) and C-terminally truncated ESAM (pcDNA3-

ESAM/ Δ 5) have been described previously [9]. The prokaryotic expression constructs encoding GST-ESAM (amino acids 278–394) and GST-ESAM Δ 21 (amino acids 278–373) were generated as described using the pGEX-KG vector [9,38]. For construction of GST-MAGI-1_{937–1035} (amino acids 937–1035 of mouse MAGI-1), the respective fragment was PCR amplified from bEnd.3 cDNA using the forward primer 5'-CCTCGAGGGCAGCGGTGTGGT-GAGCGC-3' and the reverse primer 5'-CCAAAGCT-TAAGTGTAGAAATCTTGCTCCTG-3', restricted with *Xho*I and *Hind*III, and cloned into the corresponding restriction sites of pGEX-KG. To obtain MBP-MAGI-1_{937–1035}, the *Xho*I–*Hind*III fragment from pGEX-MAGI-1_{937–1035} was subcloned into pMal-c2 (New England Biolabs, Frankfurt, Germany) using the compatible *Sal*I and *Hind*III restriction sites of the vector.

Two-hybrid screen

Two-hybrid screening experiments were performed essentially as described [39]. Briefly, the *Saccharomyces cerevisiae* reporter strain L40 expressing a fusion protein between LexA and the cytoplasmic tail of ESAM (amino acids 275–394) was transformed with 200 μ g of DNA derived from a day 9.5/10.5 mouse embryo cDNA library [40] according to the method of Schiestl and Gietz [41]. The transformants were grown for 16 h in liquid-selective medium lacking tryptophan, leucine, and uracil to maintain selection for the bait and the library plasmid. The transformants were then plated onto synthetic medium lacking tryptophan, histidine, uracil, leucine, and lysine in the presence of 1 mM 3-aminotriazole (3-AT). After 3 days at 30°C, large colonies were transferred to new plates and grown for additional 3 days on the same selective medium containing higher concentrations (15 mM) of 3-AT. Colonies growing with 15 mM 3-AT were transferred to liquid selective medium, and DNA was isolated by lysing the cells in yeast lysis solution (10 mM Tris-HCl, pH 8.0, 2% Triton X-100, 1% SDS, 100 mM NaCl, and 1 mM EDTA) in the presence of acid-washed glass beads and subsequent phenol/chloroform extraction and ethanol precipitation. To segregate the bait plasmid from the library plasmid, yeast DNA was transformed into *Escherichia coli* HB101, and the transformants were grown on M9 minimal medium lacking leucine.

In vitro binding of ESAM to MAGI-1_{792–901}

GST-ESAM fusion proteins were expressed in *E. coli* BL21 (Amersham Biosciences). The bacteria were grown at 30°C in LB medium and 2% glucose to A₆₀₀ = 0.5. Expression of recombinant proteins was induced with 1 mM isopropyl-D-thiogalactopyranoside for 1 h. The bacteria were harvested by centrifugation, resuspended in 25 ml of lysis buffer (phosphate-buffered saline, 1 “complete” protease inhibitor tablet including EDTA (Amersham Biosciences) per 25 ml lysis buffer), and lysed by passing through

a French pressure cell. GST fusion proteins were isolated by affinity chromatography on glutathione-Sepharose 4B beads (Amersham Biosciences). Purified fusion proteins were analyzed by SDS-PAGE followed by Coomassie Brilliant Blue staining. For GST pull-down experiments, the prey proteins were generated in vitro using the TNT Quick reticulocyte lysate system (Promega Corp., Madison, WI) in the presence of [35 S]methionine as described by the manufacturer. Aliquots of the translation reactions were incubated with the respective GST fusion proteins immobilized on glutathione-Sepharose 4B beads for 1 h at 4°C under constant agitation. Beads were washed five times with buffer containing 10 mM HEPES, pH 7.4, 100 mM KCl, 1 mM MgCl₂, and 0.1% Triton X-100. Bound proteins were eluted by boiling for 5 min in SDS sample buffer, subjected to SDS-PAGE, and analyzed by fluorography.

Antibodies

To raise rabbit antibodies against mouse MAGI-1, GST-MAGI-1_{937–1035} (GenBank™ accession number of MAGI-1b: AF027503) was expressed in *E. coli*, isolated by affinity chromatography on glutathione-Sepharose 4B beads (Sigma), and injected into New Zealand White rabbits. Antiserum (VE12) was affinity-purified using MAGI-1_{937–1035} fused to maltose-binding protein (MBP-MAGI-1_{937–1035}) and immobilized on CNBr-Sepharose (Amersham Biosciences). Monoclonal rat anti-ESAM antibodies were described previously [9]. The following antibodies were commercially obtained: mouse anti-FLAG monoclonal antibody M2 (Sigma); rabbit polyclonal antibody ANTI-FLAG (Sigma); rabbit polyclonal antibody against ZO-1 (Zymed Laboratories Inc., San Francisco, CA); secondary antibodies, and fluorochrome-conjugated reagents (Dianova, Hamburg, Germany).

Immunoprecipitation

CHO transfectants stably expressing full-length ESAM (CHO-ESAM) [9] were transiently transfected with MAGI-1-FLAG or a mock construct using GeneJammer reagent (Stratagene) according to the manufacturers protocols. In control experiments, nontransfected CHO cells transiently transfected with MAGI-1-FLAG were subjected to immunoprecipitation. Confluent monolayers of these cells were rinsed three times with ice-cold phosphate-buffered saline (PBS), supplemented with 1 mM MgCl₂ and 1 mM CaCl₂ (PBSMC). Cells were harvested in ice-cold PBS supplemented with 2 mM dithiothreitol and 1 “complete” protease inhibitor tablet, EDTA-free (Amersham Biosciences) per 25 ml buffer, and sedimented by centrifugation. The supernatant was discarded and the cells were extracted overnight under constant agitation in lysis buffer containing 20 mM imidazole, pH 6.8, 100 mM NaCl, 5 mM EDTA, 1 mM dithiothreitol, 0.5% Triton X-100, and 1 “complete” protease inhibitor tablet (EDTA-free) per 50 ml buffer. The lysates

were clarified by centrifugation at 30,000 × *g* for 30 min at 4°C. The cell lysates were incubated overnight with 30 µl protein G-Sepharose beads and 10 µg anti-ESAM mAb 1G8 or 10 µg isotype-specific control antibody or 10 µg anti-FLAG mAb (M2). Immunocomplexes were washed four times with lysis buffer (protease inhibitor tablet omitted), subjected to SDS-PAGE, and analyzed by immunoblotting with affinity-purified anti-MAGI-1 antibodies (VE12). Bound antibodies were visualized with horseradish peroxidase-conjugated donkey anti-rabbit IgG (Dianova) by ECL.

Immunofluorescence of HUVEC, tissue sections, and transfected CHO cells

HUVECs were plated on polycarbonate filters (Transwell™, 0.4-µm pore size, 6.5-mm diameter, from Costar, Corning, NY) coated with gelatin (Sigma). The filters were rinsed three times with PBSMC and subsequently fixed in methanol for 10 min at –20°C. After rehydration in PBS, the cells were blocked with 1% (w/v) bovine serum albumin in PBS to minimize unspecific binding. The cells were incubated with primary antibodies for 1 h, and with secondary antibody for 30 min in blocking buffer at room temperature, followed by three times washing with phosphate-buffered saline after each incubation. Finally, the cells were mounted with fluorescent mounting medium (DAKO, Hamburg, Germany) and viewed with an Axioskop 2 (Zeiss, Oberkochen, Germany) fluorescence microscope. Controls included incubation of fixed cells with purified MAGI-1 preimmune serum. For immunofluorescence analysis of ectopically expressed MAGI-1αβ2γ in CHO, CHO-ESAM, or CHO-ESAMΔ5 cells, cells were grown on LabTec chamber slides (Nalge-Nunc, Wiesbaden, Germany), transiently transfected with MAGI-1-FLAG or a mock construct using GeneJammer reagent (Stratagene), and then stained as described above. For immunofluorescence staining of mouse tissue, the tongues of Balb/c mice were embedded in Tissue Tek OCT compound (Miles, Elkhart, IN), snap frozen, and stored at –80°C. Sections of 8 µm were cut on a freezing microtome, mounted on slides, coated with poly-L-lysine (Menzel-Gläser, Nußloch, Germany), and dried. Sections were then fixed with 4% (w/v) paraformaldehyde in PBS for 20 min at room temperature, permeabilized with PBS containing 0.1% (w/v) Triton X-100 for 2 min at 4°C, and blocked with PBS containing 5% (w/v) bovine serum albumin for 30 min, followed by an additional blocking step with 20% (v/v) goat serum in PBS for 1 h at room temperature. The primary and secondary antibodies were diluted with PBS containing 10% (v/v) goat serum. The primary antibodies were incubated overnight at 4°C, followed by PBS containing 0.1% (v/v) Triton X-100 and PBS containing 5% (w/v) bovine serum albumin for 30 min at room temperature. The secondary antibodies were incubated for 1 h at room temperature. All incubation steps, except the blocking steps, were followed by extensive washing with PBS. Finally, the sections were mounted with

fluorescent mounting medium (DAKO) and viewed with a TCS confocal imaging system (Leica, Bensheim, Germany). To visualize the primary antibodies by immunofluorescence, Cy3-conjugated anti-rabbit and Cy2-conjugated anti-rat secondary antibodies were used to detect anti-MAGI-1 pAb VE12 or anti-ESAM mAb 1G8, respectively. Control experiments included single immunostainings and specificity tests for secondary antibody binding.

Accession numbers

The nucleotide sequence for mouse MAGI-1 $\alpha\beta\gamma$ has been deposited in the GenBank database under GenBank accession number AY497557.

Abbreviations

The abbreviations used are Ig, immunoglobulin; CHO, Chinese hamster ovary; HUVEC, human umbilical vein endothelial cells; CTX, cortical thymocyte marker of *Xenopus*; ECL, enhanced chemiluminescence; PAGE, polyacrylamide gel electrophoresis; PBS, phosphate-buffered saline; SDS, sodium dodecyl sulfate; GST, glutathione S-transferase; mAb, monoclonal antibody; pAb, polyclonal antibody.

Results

Identification of MAGI-1 as a protein interacting with the cytoplasmic domain of ESAM

To identify proteins interacting with ESAM, we screened a murine embryonic cDNA library [40] by yeast two-hybrid selection with the full-length cytoplasmic domain of ESAM as bait. A cDNA fragment was isolated that corresponds to nucleotides 3298–3627 of the mouse MAGI-1 cDNA sequence [24]. The fragment is present in all three originally identified splice variants of mouse MAGI-1: MAGI-1a, MAGI-1b, and MAGI-1c (GenBank™ accession number of MAGI-1b: AF027503). It encodes amino acids 792–901 of MAGI-1 (MAGI-1_{792–901}) harboring the PDZ3 domain of this protein that extends from amino acid residues 789–876 (Fig. 1A). ESAM possesses a type I consensus motif for binding to PDZ domains at its very C-terminus (type I motif: -S/T-X- Φ ; where Φ is a hydrophobic amino acid and X is any amino acid). Thus, this finding suggests that the interaction between ESAM and MAGI-1 is PDZ domain mediated. No other MAGI-1 clone was obtained in the screen using the cytoplasmic domain of ESAM as bait.

In vitro binding of ESAM to MAGI-1

To confirm the association between ESAM and MAGI-1 biochemically, we performed GST pull-down assays. The full-length cytoplasmic tail of ESAM fused to GST (GST-

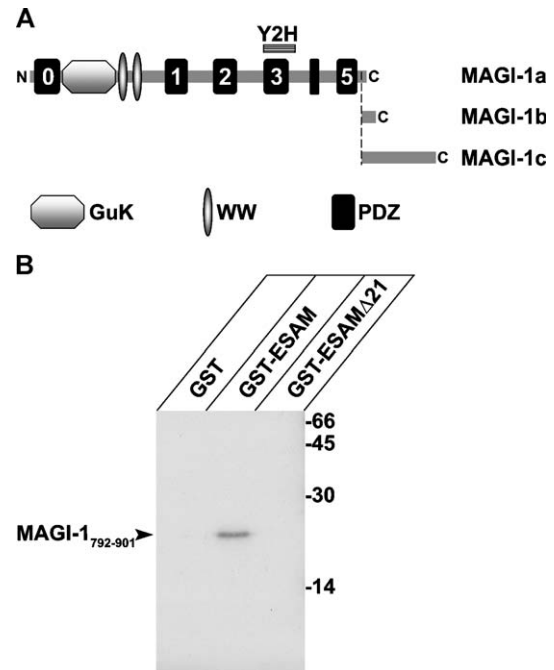


Fig. 1. MAGI-1 interacts with the cytoplasmic domain of ESAM. (A) Schematic domain organization of the C-terminal splice variants of mouse MAGI-1: MAGI-1a, MAGI-1b, and MAGI-1c. The different domains are drawn as indicated below. Alternative splicing of MAGI-1 within the PDZ4 region of the originally identified splice variants of mouse MAGI-1 (MAGI-1a, MAGI-1b, and MAGI-1c) is not recognized as an intact PDZ domain by several protein domain databases (data not shown). Therefore, PDZ4 is replaced by a shortened box. The hatched bar (Y2H) indicates the MAGI-1 fragment (amino acids 792–901) isolated from the yeast two-hybrid library using the cytoplasmic domain of ESAM as bait. (B) Interaction of GST-ESAM with the MAGI-1 fragment obtained in the yeast two-hybrid screen. The MAGI-1 fragment isolated from the yeast two-hybrid library (MAGI-1_{792–901}) was generated in vitro in the presence of [³⁵S]methionine and incubated with immobilized GST fusion proteins containing the complete cytoplasmic tail of ESAM (GST-ESAM), the cytoplasmic tail lacking 21 C-terminal amino acid residues (GST-ESAMΔ21), or with GST alone (GST). Specifically bound proteins were eluted, electrophoresed, and visualized by fluorography. The labeled MAGI-1_{792–901} fragment (position indicated by an arrowhead) comprising the PDZ3 domain of MAGI-1 interacts with ESAM most likely in a PDZ-dependent manner. Molecular mass markers (in kDa) are indicated on the right.

ESAM), the truncated cytoplasmic domain of ESAM lacking the last 21 C-terminal amino acids (GST-ESAMΔ21), or GST alone was immobilized on glutathione resin and incubated with in vitro translated and [³⁵S] methionine-labeled recombinant protein fragments comprising the PDZ3 domain of MAGI-1 (MAGI-1_{792–901}). MAGI-1_{792–901} binding was observed with GST-ESAM (Fig. 1B). The interaction was abolished upon truncating the C-terminus of ESAM (GST-ESAMΔ21). No MAGI-1_{792–901} fragment was captured in pull-down assays using GST alone (Fig. 1B). This demonstrates that the association between MAGI-1 and ESAM was specific. These results indicate that ESAM directly interacts with the PDZ3 domain of MAGI-1 through its C-terminal tail, most likely through its PDZ binding motif.

Identification of a novel splice variant of MAGI-1c

To further analyze the association of ESAM with MAGI-1, we cloned full-length mouse MAGI-1c from bEnd.3 cells by reverse transcription-polymerase chain reaction. Based on the published MAGI-1 sequence, we expected an amplification product of 4130 nucleotides [24]. However, we obtained an amplification product of 4416 bp. Cloning and sequencing revealed a new splice variant of mouse MAGI-1c that is 96 amino acids longer than the originally described form. It contains three insertions of which two, the α - and the β 2-insertion, have been previously identified in mouse and in human MAGI-1 [26,30]. The α -insertion is a stretch of 28 amino acids located N-terminally of the PDZ3 domain. The β 2-region is a stretch of 56 amino acids that is inserted C-terminally of the partial PDZ4 domain, completing it to PDZ4. In addition, we identified a third domain that is alternatively spliced in MAGI-1c. This domain is an inserted region of 12 amino acids preceding the second WW domain. It could be identical to an insertion in MAGI-1 mentioned in an earlier study [30], although detailed information was not provided in this report. Following the nomenclature for human MAGI-1 splice variants proposed by Laura et al. [26], we named this insertion the γ -insert of MAGI-1c and the new splice variant containing all three insertions we propose to name MAGI-1c $\alpha\beta$ 2 γ . Database searches revealed the presence of these insertions in overlapping expressed sequence tags (ESTs). Fig. 2A summarizes the

domain organization of the previously described mouse MAGI-1c molecule [24] and the newly identified splice variant MAGI-1c $\alpha\beta$ 2 γ . Fig. 2B displays sequence alignments of the two molecules. Both MAGI-1c forms contain MAGI-1_{792–901}, the region harboring PDZ3 that corresponds to the ESAM-interacting fragment we isolated in the yeast two-hybrid screen. We used the newly identified splice variant MAGI-1c $\alpha\beta$ 2 γ to further examine the association between ESAM and MAGI-1. To be able to detect MAGI-1 in various immunoassays, we attached a FLAG tag to the C-terminus of MAGI-1c $\alpha\beta$ 2 γ (for brevity, called MAGI-1-FLAG). Additionally, we generated polyclonal antibodies against a distinct region of MAGI-1c extending from amino acid residues 937–1035 (see open bar in Fig. 2A) that is also included in MAGI-1a and MAGI-1b splice variants, but shows no significant sequence similarity to MAGI-2 or MAGI-3 sequences. We evaluated the ability of the antiserum to detect MAGI-1 in immunoblots of various cell extracts. First, CHO cells stably expressing ESAM (CHO-ESAM) [9] were transiently transfected with MAGI-1-FLAG (CHO-ESAM-M). Subsequently, lysates were analyzed with affinity-purified anti-MAGI-1 (VE12). The antibody recognized MAGI-1-FLAG (Fig. 3A). The recombinant protein migrated with an apparent molecular mass of about 185 kDa. No immunoreactive bands were seen in control experiments using CHO-ESAM cells, lacking transfected MAGI-1. A very weak band of similar apparent molecular weight as MAGI-1 was detected if more than 10 times larger

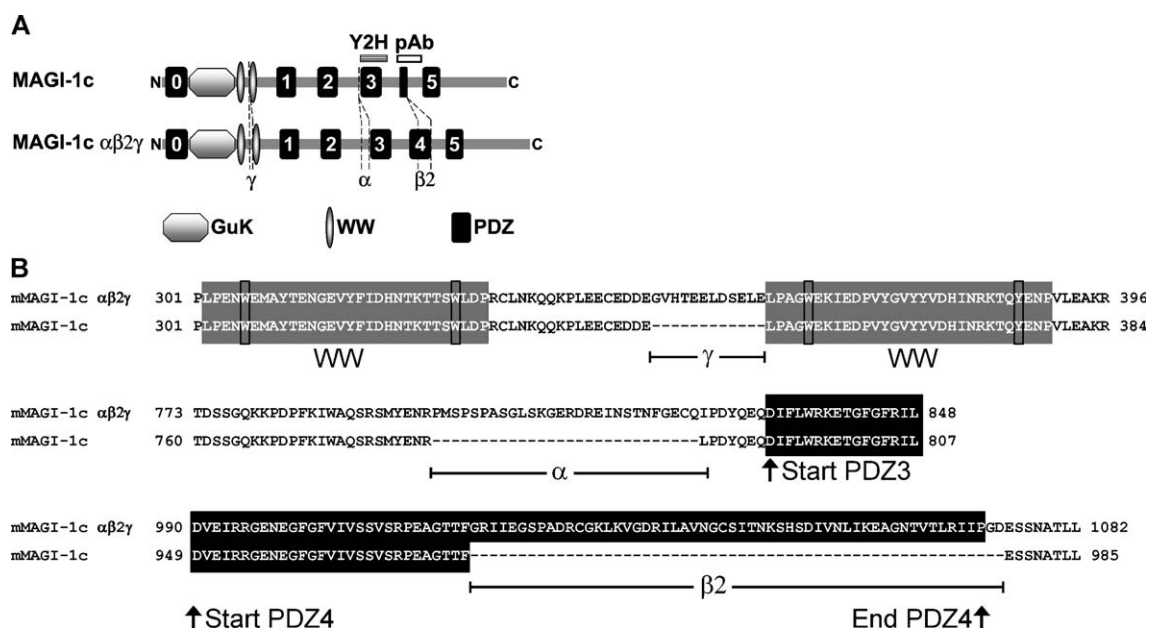


Fig. 2. Organization of the newly identified MAGI-1c splice variant (MAGI-1c $\alpha\beta$ 2 γ) compared to the database sequence of mouse MAGI-1c. (A) Schematic domain organization of MAGI-1c. Dashed lines display the alternatively spliced sequences. The hatched bar (Y2H) indicates the MAGI-1 fragment (amino acids 792–901) isolated from the yeast two-hybrid library, the open bar (pAb) represents the immunogen sequence used for the generation of polyclonal anti-MAGI-1 specific antibodies. (B) Sequence alignment of the alternatively spliced sequences. The WW domain sequences are displayed in gray, the PDZ domain sequences in black. The typical hydrophobic amino acid residues of the WW domains are boxed. Solid bars underneath the alignment represent the spliced in-out sequences. GenBank™ accession number of mouse MAGI-1c: AF027505.

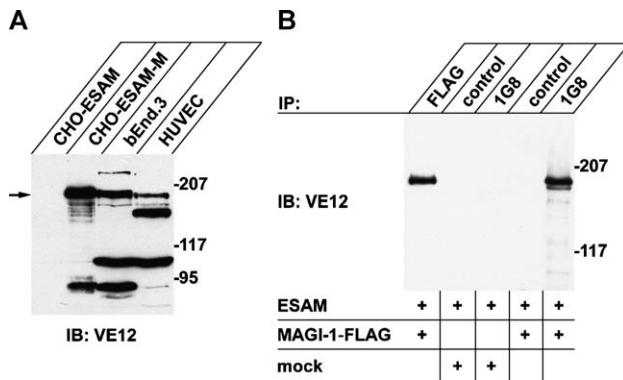


Fig. 3. Characterization of anti-MAGI-1 antibodies and association of ESAM with MAGI-1 in mammalian cells. (A) Immunoblot (IB) analysis of MAGI-1 in transfected Chinese hamster ovary cells (CHO cells), bEnd.3 endothelioma cells, and primary human umbilical vein endothelial cells (HUVEC). Urea/SDS extracts of the respective cell type were subjected to SDS-PAGE, transferred to filters, and immunoblotted with affinity-purified anti-MAGI-1 (VE12). CHO cells stably expressing ESAM (CHO-ESAM) were left untransfected (lane 1) or were transfected with MAGI-1-FLAG (CHO-ESAM-M, lane 2). VE12 recognized a 185-kDa protein (lane 2) that was not detectable in nontransfected cells (lane 1), indicating that the antibody is capable of recognizing recombinant MAGI-1. In HUVEC or bEnd.3 cell extracts, a band migrating at the same relative molecular weight as MAGI-1-FLAG can be observed. VE12 recognizes additional bands in all analyzed cell types that probably correspond to different MAGI-1 splice variants, although protein degradation cannot be ruled out. The arrow indicates the position of MAGI-1-FLAG in lane CHO-ESAM-M. Molecular mass markers (in kDa) are indicated on the right. (B) Co-immunoprecipitation (IP) of ESAM and MAGI-1. CHO-ESAM cells were transfected with MAGI-1-FLAG or with an irrelevant construct (mock), as indicated below. Lysates were incubated with anti-FLAG mAb, anti-ESAM mAb (1G8), or with an isotype-specific control antibody, as indicated above. Immunoprecipitates were analyzed by SDS-PAGE and immunoblotting with affinity-purified anti-MAGI-1 (VE12). The protein load in lane 1 (FLAG) was much reduced compared to the other lanes. Molecular mass markers are indicated on the right.

quantities of cell extracts of untransfected CHO-ESAM cells were analyzed (data not shown). Thus, we cannot rule out that CHO cells may express very weak levels of endogenous MAGI-1. Interestingly, immunoblot analysis of MAGI-1 in CHO cells that were only transfected with MAGI-1 but not with ESAM revealed strong degradation of MAGI-1, suggesting that ESAM may stabilize MAGI-1 and protect it from degradation (data not shown). We next asked whether VE12 would recognize MAGI-1 in endothelial cells that are well known for expressing the binding partner ESAM [9,10]. Immunoblot analysis of mouse endothelioma cells (bEnd.3) and human umbilical vein endothelial cells (HUVEC) showed several bands, with two major ones of approximately 185 and 105 kDa and a weaker one of approximately 165 kDa in both cell extracts (Fig. 3A). In HUVEC, a prominent band of approximately 150 kDa, and in bEnd.3 cells, a prominent band of approximately 90 kDa, were also detected with VE12. No signals were seen when control blots were incubated with peroxidase-conjugated secondary antibodies alone (data not shown). Comparison of the 185-kDa bands we observed in endothelial cell extracts with the band obtained for MAGI-

1-FLAG showed that the proteins comigrated. This suggests that the 185-kDa bands observed in endothelial cells might correspond to MAGI-1 $\alpha\beta\gamma$ or a very similar splice variant. The protein bands of lower apparent molecular mass may represent additional splice variants of MAGI-1 or protein degradation. Taken together, affinity-purified VE12 was capable of recognizing recombinant MAGI-1. Immunoblotting of endothelial cell extracts with the same antibody uncovered cell type-specific sets of bands that most likely represent MAGI-1 splice variants.

ESAM interacts with the newly identified MAGI-1c splice variant in mammalian cells

To further confirm the interaction between ESAM and MAGI-1, we first attempted to co-immunoprecipitate ESAM and MAGI-1 from endothelioma cells. However, we could not demonstrate an association of the endogenous proteins by standard immunoprecipitation experiments. This might be explained by a weak interaction of the two binding partners; in addition, it is also possible that, in contrast to ESAM, MAGI-1 is highly resistant to Triton X-100 extraction as already shown for MAGI-1 in kidney glomeruli [32]. Therefore, we overexpressed MAGI-1-FLAG or a mock construct in CHO transfectants stably expressing ESAM (CHO-ESAM). Cell extracts were incubated with anti-ESAM mAb 1G8 [9], isotype-specific control antibody, or anti-FLAG mAb. The precipitated material was analyzed in immunoblots for the presence of MAGI-1-FLAG using the anti-MAGI-1 antibody VE12. The anti-ESAM mAb 1G8 coprecipitated the MAGI-1 splice variant (Fig. 3B). Use of the isotype-specific control antibody did not result in MAGI-1 precipitation. No bands were observed when mock-transfected CHO-ESAM cells or CHO cells ectopically expressing the MAGI-1 splice variant alone were subjected to immunoprecipitation experiments (Fig. 3B and data not shown). To confirm the integrity and the detergent solubility of expressed MAGI-1-FLAG in CHO-ESAM cells, immunoprecipitations with anti-FLAG mAb were performed in parallel experiments (Fig. 3B). Taken together, these findings indicate that ESAM and MAGI-1 associate in mammalian cells. Thus, three independent methods, yeast two-hybrid selection, GST pull-down, and co-immunoprecipitation show that MAGI-1 is a binding partner for ESAM.

MAGI-1 is localized in endothelial cell contacts

We next examined the distribution of MAGI-1 in endothelial cells by indirect immunofluorescence. Monolayers of filter-grown human umbilical vein endothelial cells (HUVEC) were stained with antibodies to MAGI-1 (VE12) or ZO-1 or with MAGI-1 preimmune serum. MAGI-1 was localized at sites of cell-cell contact (Fig.

4A). The image showed intensely labeled spots giving a rim-like staining at the periphery of the cells in addition to some immunoreactivity in the cytoplasm. Observed labeling was specific as evidenced by the lack of staining obtained with MAGI-1 preimmune IgG. Anti-ZO-1 was used to visualize the cell contact areas formed by HUVEC. The MAGI-1 staining was reminiscent of the pattern observed with antibodies against ZO-1 (Fig. 4A). Thus, a substantial fraction of MAGI-1 resided at cell–cell contact sites. To investigate the potential colocalization of ESAM and MAGI-1 in situ, we analyzed the localization of the molecules in cryosections of mouse tongue by confocal immunofluorescence microscopy. Double labeling with antibodies to ESAM (1G8) and MAGI-1 (VE12) clearly revealed colocalization of the proteins at interendothelial cell contacts (Fig. 4B). Interestingly, MAGI-1 seemed to be present also at the luminal surface of endothelial cells.

ESAM-dependent recruitment of MAGI-1 at sites of cell–cell contact

The subcellular localization of MAGI-1 in HUVEC and in mouse endothelium prompted us to test whether ESAM would recruit MAGI-1 to sites of cell–cell contact. To address this question, we transiently expressed MAGI-1-FLAG in CHO-ESAM, CHO-ESAM Δ 5 cells, or nontransfected CHO cells, and analyzed the distribution of the MAGI-1 splice variant in these cells by double-label immunofluorescence (Fig. 5). In nontransfected CHO cells lacking ESAM, MAGI-1-FLAG displayed a cytoplasmic distribution and was not enriched at sites of cell–cell contact (Fig. 5, top panel). In CHO-ESAM cells, ESAM is localized to areas of cell–cell contact most likely by way of homophilic binding (Fig. 5, middle panels) [9]. When CHO-ESAM cells were transiently transfected with MAGI-1-FLAG, the protein colocalized with ESAM at cell–cell contacts (Fig. 5, middle panel). In

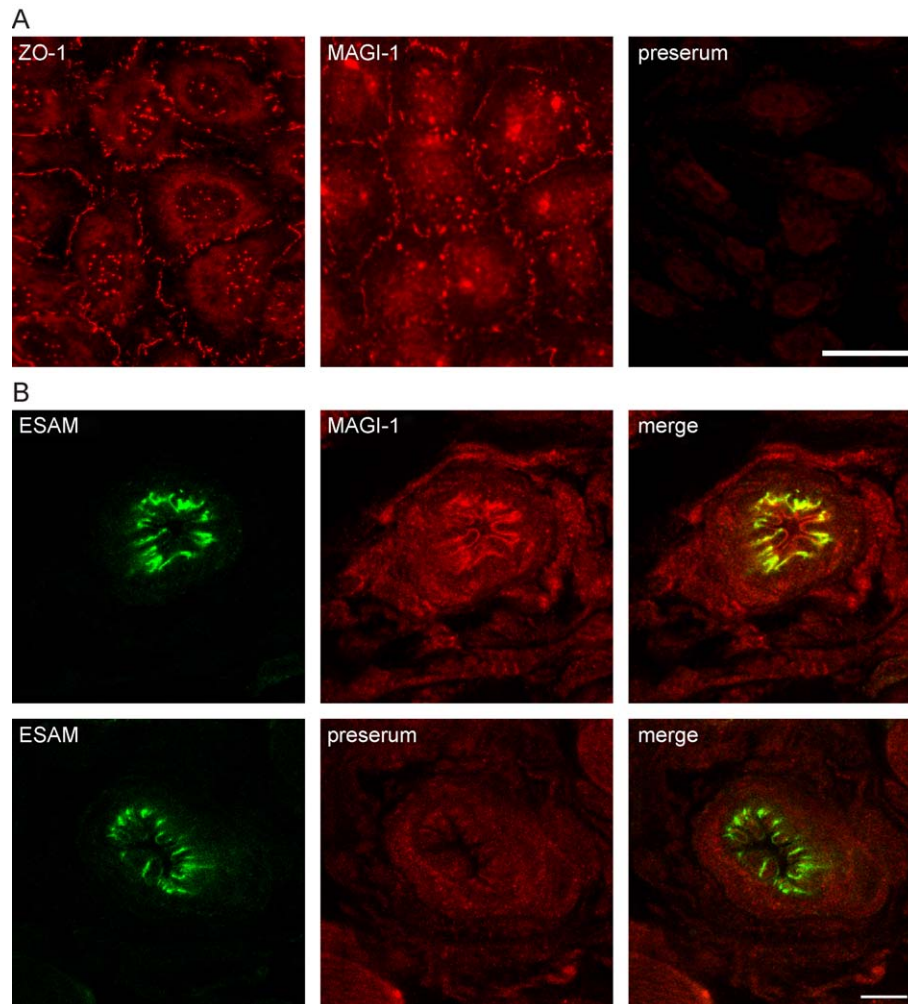


Fig. 4. MAGI-1 localizes at cell–cell contacts of endothelial cells. (A) Human umbilical vein endothelial cells (HUVEC) were stained with antibodies to ZO-1 or MAGI-1 (VE12) or with purified MAGI-1 preimmune IgG (presum). Bound antibodies were visualized with Cy3-conjugated secondary antibodies. MAGI-1 localizes at cell–cell contacts of HUVECs in a similar way to ZO-1. Scale bar: 50 μ m. (B) Tissue sections of mouse tongue were double-stained by indirect immunofluorescence for ESAM (green) and MAGI-1 (red) using the rat mAb 1G8 and the rabbit pAb VE12, respectively. MAGI-1 clearly colocalizes with ESAM at interendothelial cell contacts and is also located at the luminal surface of endothelial cells. Scale bar: 8 μ m.

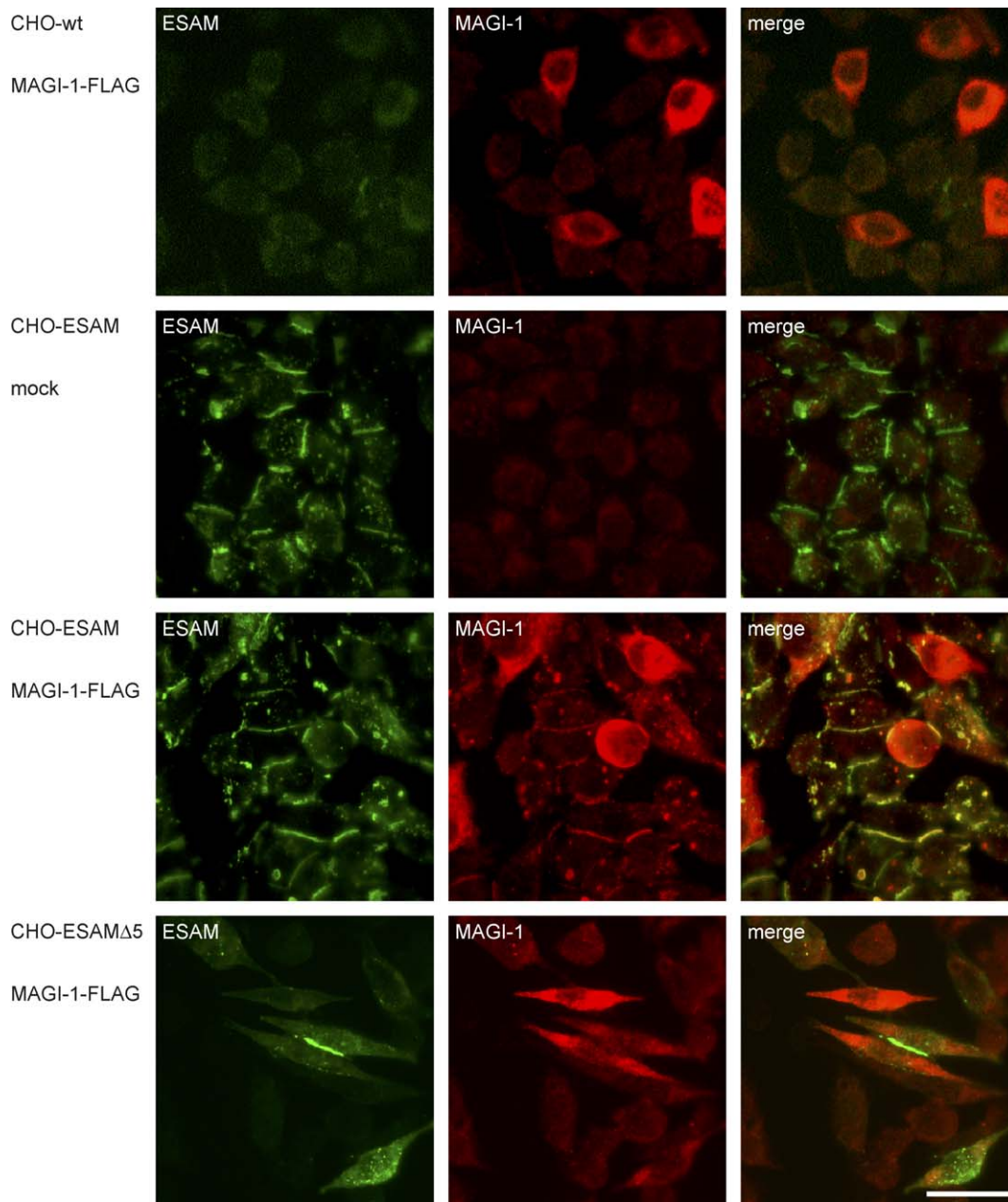


Fig. 5. ESAM recruits MAGI-1 to cell–cell contacts in transfected CHO cells. CHO cells stably transfected with ESAM (CHO-ESAM), C-terminally truncated ESAM (CHO-ESAM Δ 5), or nontransfected CHO (CHO-wt) cells were transiently transfected with MAGI-1-FLAG or with an irrelevant construct (mock), as indicated on the left. CHO-ESAM and CHO-wt cells were double stained by indirect immunofluorescence for ESAM (green) and for MAGI-1 (red) using the rat mAb 1G8 and the rabbit pAb VE12, respectively. In CHO-ESAM Δ 5 cells, MAGI-1-FLAG was detected using the rabbit pAb ANTI-FLAG (Sigma). As depicted in the merged images, MAGI-1-FLAG clearly colocalizes with ESAM. In co-transfected cells, the MAGI-1c splice variant is clustered at cell–cell contacts, whereas in cells lacking ESAM or expressing the C-terminally truncated molecule (ESAM Δ 5), it is mainly localized in the cytoplasm. Scale bar: 50 μ m.

CHO-ESAM Δ 5 cells, ESAM lacking the last five C-terminal amino acids was targeted at sites of cell–cell contact just like full-length ESAM. When CHO-ESAM Δ 5 cells were transiently transfected with MAGI-1-FLAG, it was not concentrated at cell contact sites, it showed a cytoplasmic distribution (Fig. 5, bottom panel). We obtained these results by staining the cells simultaneously with anti-ESAM mAb (1G8) and anti-MAGI-1 (VE12) (Fig. 5) or with 1G8 and

anti-FLAG mAb (data not shown). In CHO-ESAM Δ 5 cells, MAGI-1-FLAG was detected by the pAb ANTI-FLAG. VE12 occasionally stained cell–cell contact sites of CHO-ESAM cells very weakly in the absence of MAGI-1-FLAG, suggesting some very weak endogenous expression of MAGI-1, which would be in agreement with similarly weak signals for endogenous MAGI-1 in immunoblots (data not shown). Control experiments included single immunostain-

ings, specificity tests for secondary antibody binding, and mock-transfections of CHO-ESAM cells (Fig. 5, middle panel and data not shown). Taken together, our findings suggest that ESAM recruits MAGI-1 to sites of cell–cell contact in CHO cells. The recruitment of MAGI-1 is dependent on the PDZ domain-binding motif of ESAM.

Discussion

ESAM is an immunoglobulin-like transmembrane protein associated with endothelial tight junctions. We attempted to gain more insight into the molecular function of ESAM by searching for potential intracellular binding partners. In the current study, we have identified the membrane-associated guanylate kinase protein MAGI-1 as an ESAM-interacting molecule. GST-ESAM associated with a MAGI-1 fragment consisting almost exclusively of the PDZ3 domain of MAGI-1; deletion of the 21 C-terminal amino acids of GST-ESAM abolished the binding. These findings strongly suggest a direct association between ESAM and MAGI-1 through a PDZ domain-mediated interaction. ESAM and MAGI-1 can be co-immunoprecipitated, indicating that the association is relevant *in vivo*. Upon ectopic expression in CHO cells stably expressing ESAM, MAGI-1 colocalized with ESAM at cell–cell contacts, indicating that ESAM can recruit MAGI-1. The recruitment is PDZ domain dependent as an ESAM mutant lacking the last five C-terminal amino acids did not support localization of MAGI-1 at cell–cell contacts. Our results define MAGI-1 as the first cytosolic protein that directly interacts with ESAM and can use ESAM as an anchor at the cytoplasmic surface of cell–cell contact sites.

We have recently found that not only JAM-A, but also JAM-B and JAM-C bind to the tight junction-associated PDZ domain proteins ZO-1 and PAR-3 [16]. In contrast, neither ESAM nor CAR bound to any of these two PDZ domain proteins. With MAGI-1, we have now found the first PDZ domain protein that binds to ESAM. This adaptor protein was also found to bind to yet another junction-associated CTX family member JAM4, but not to JAM-A [32]. In combination with these findings, our results suggest that there are two subfamilies of tight junction-associated members of the CTX family. One formed by JAM-A, -B, and -C, the other formed by ESAM and JAM4 based on the type of PDZ domain proteins they bind to. These different specificities may allow these members of the two different groups of IgSF molecules to interact with different multi-protein complexes and thus to interact with different signaling pathways or cytoskeletal elements at tight junctions. This proposal of two different groups of tight junction-associated IgSF members is in good agreement with the analysis of the sequence similarity of the different CTX members. As shown in the phylogenetic tree in Fig. 6, ESAM, CAR, CTM, JAM4, and BT-IgSF (forming the CTX subfamily) are clearly more closely related to each other

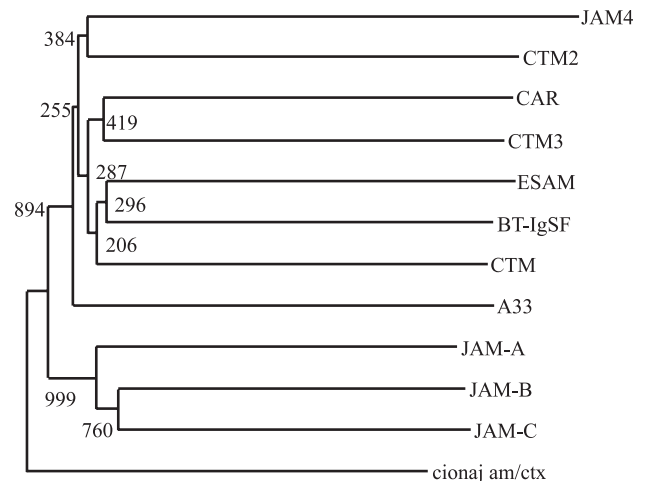


Fig. 6. Phylogenetic tree indicating the relationships among the CTX family members (mouse sequences). The CTX family members cluster in two separate groups, the CTX subfamily and the JAM subfamily. Alignments were performed with the Clustal W program and the tree obtained with the neighbor joining algorithm offered by the Clustal package [51]. Bootstrap values for the delineation of the two subfamilies (999 for the JAM subfamily and 894 for the CTX subfamily) are highly significant. A *Ciona* gene corresponding to a preduplication form of the CTX family [52] was added as an outgroup. Accession to the various sequences: CTM: NM_020518; CTM2: NP_084457; CTM3 (is identical to CLMP): NP_598494; BT-IgSF: NM_170599; CAR: BC016457; A33: NM_021610; ESAM: AF361882; JAM-A: MMU89915; JAM-B: NM_023277; JAM-C: NM_023844; JAM4: BC004806 AF537215.

than to JAM-A, -B, and -C (forming the JAM subfamily). Furthermore, in agreement to this, the latter group shares a class-II PDZ domain-binding motif while each of the former carries a class-I PDZ binding motif. In addition, the cytoplasmic tails of the former group are much longer than those of the latter. Future studies will clarify whether these two groups of tight junction-associated CTX members indeed fulfil different functions at tight junctions. In a very recent study by Raschperger et al. [12], analysis of the phylogenetic relationships between members of the CTX family also revealed the formation of subfamilies or groups within this family of IgSF members. Similar to our findings, the authors propose a JAM group comprised of JAM-A, -B, and -C that was phylogenetically separated from a group consisting of CLMP, CAR, ESAM, and BT-IgSF which they named CAR group. In addition, Raschperger et al. defined a third group within the CTX family, the CTX group containing CTM and A33. The authors showed a close relationship of the CTX group members to the JAM group, whereas based on our results these proteins were more closely related to the CAR-like molecules. This difference may be based on analyzing a slightly varying repertoire of CTX family proteins.

ESAM is known to be present at tight junctions in endothelial cells [9]. Up to now, MAGI-1 has also been shown to be located at tight junctions by several reports; however, these studies were limited to epithelial cells [26,42,43]. Endothelial expression of MAGI-1 has been

described based on immunohistochemical staining of MAGI-1 on endothelia of the pancreas and on the detection of MAGI-1 by immunoblots of HUVECs [26]. Our data further confirm endothelial expression of MAGI-1 because we cloned the MAGI-1 splice variant from bEnd.3 endothelioma cells and we detected MAGI-1 in immunoblots on bEnd.3 cells or HUVECs. Importantly, we show here for the first time that MAGI-1 is indeed expressed at cell contacts of endothelial cells and that it is colocalized with ESAM at these sites.

JAM4, the second CTX member that binds to MAGI-1, was found to bind preferentially to PDZ1 and PDZ4 of MAGI-1 [32]. Thus, ESAM and JAM4 appear to interact with different PDZ domains of MAGI-1. Like ESAM, JAM4 is present at tight junctions and mediates homophilic adhesion [32]. A sequential or even a simultaneous binding to MAGI-1 would be conceivable if the two proteins could coexist in the same cell. However, whether JAM4 is indeed expressed in endothelial cells is at present unknown.

In a recent report, the potential role of ESAM in angiogenesis has been investigated in transgenic mice with a gene-targeted deletion of the ESAM gene [34]. Interestingly, physiological angiogenesis was normal in ESAM-deficient mice, but ESAM seemed to have a significant role in pathological angiogenesis associated with tumor growth. In addition, *in vitro* studies on ESAM-deficient endothelial cells or a cell line that overexpressed ESAM suggested that ESAM is involved in migration and tube formation of endothelial cells [34]. By analogy with other adhesion molecules, ESAM might be implicated in cell signaling events underlying angiogenic processes. In this context, ESAM would be expected to interact with signal transducing molecules or cytoskeletal elements. MAGI-1 is indeed thought to link proteins to cytoskeletal elements [29] and appears to be involved in signal transduction processes (as discussed below). Thus, it will be interesting to elucidate whether the interaction of ESAM with MAGI-1 is of relevance for the process of tumor angiogenesis.

A possible role of MAGI-1 in angiogenesis might be suggested in another context, because MAGI-1 binds to the brain-specific angiogenesis inhibitor 1 (BAI1) [25]. BAI1 is a putative anti-angiogenic factor that might control blood vessel growth and density in the brain and could have tumor-suppressive properties [44]. It interacts with the PDZ3 or PDZ4 domain of MAGI-1 [44]. However, BAI1 seems to be absent from endothelia [45], excluding a sequential or a simultaneous binding of ESAM and BAI1 to MAGI-1.

Support for a function of MAGI-1 and its isoforms MAGI-2 and MAGI-3 in signal transduction events came from protein–protein interaction experiments. MAGI-1 can bind to RapGEP, a GDP/GTP exchange protein, and to mNET1, a Rho family nucleotide exchange factor [46,47]. Recently, MAGI-3 was found to interact with receptor tyrosine phosphatase β (RPTP β), a phosphatase that mediates cell–cell and cell–matrix interactions [48]. Two

separate studies have uncovered a functional role of MAGI-2 and MAGI-3 [42,49]. Both proteins are present at tight junctions and associate with the regulatory molecule phosphatase and tensin homolog (PTEN). This interaction has been proposed to recruit PTEN to tight junctions. PTEN attenuates signaling via the phosphoinositide 3-kinase pathway and is thereby a negative regulator of protein kinase B/Akt activity. PTEN signaling via protein kinase B/Akt appears to suppress cell division and enables apoptosis [50]. The interaction between PTEN and MAGI-2 or MAGI-3 is PDZ2 domain mediated. Interestingly, both MAGIs seem to cooperate with PTEN to modulate the kinase activity of protein kinase B/Akt. Binding to the MAGI isoforms enhances the ability of PTEN to suppress protein kinase B/Akt activation [42,49]. In view of a potential role for ESAM in signaling, one might speculate that ESAM recruits PTEN to endothelial tight junctions via MAGI-1, provided that PTEN can also interact with MAGI-1. This is conceivable due to the high homology between the MAGI isoforms, but remains to be shown. Alternatively, ESAM might target PTEN to tight junctions through interaction with MAGI-2 or MAGI-3, assuming that ESAM is able to bind to MAGI-2 or MAGI-3 and that MAGI-2 or MAGI-3 is also expressed by endothelial cells.

In summary, we have identified MAGI-1 as a protein that binds directly to the cytoplasmic domain of ESAM in a PDZ domain-dependent manner. Furthermore, our data suggest that ESAM can trigger recruitment of MAGI-1 to intercellular contacts. Thus, the association of ESAM with MAGI-1 might allow the assembly of multiprotein complexes, in which ESAM might serve as an anchor to recruit appropriate signaling and cytoskeletal proteins to endothelial tight junctions.

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